

Engineering Architecture Overview

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System Overview

The Monk-E platform is composed of API services, asynchronous workers, a relational system of record, object storage, and a retrieval layer for internal knowledge applications.

Services communicate using authenticated HTTP APIs and managed background jobs. Configuration is supplied through environment-specific deployment configuration rather than committed secrets.

Data Flow

User requests enter through the API gateway, pass identity and policy checks, and are routed to the relevant application service. Knowledge retrieval uses indexed chunks and metadata filters.

Document ingestion is asynchronous for larger files. Original files remain in object storage, while normalized text, metadata, and embeddings are indexed for search.

Reliability Principles

Stateless application workers are preferred. Long-running or retryable work should be represented as jobs with explicit idempotency keys.

Critical operations emit structured logs and metrics. Dependency failures should degrade predictably rather than returning partial or misleading business results.

Security Boundaries

Authentication establishes identity; authorization decides access. Sensitive business data should never be injected into an LLM context unless the request has passed the same authorization boundary as the underlying source.

Synthetic prototype corpus - not real company data.